

Electrodynamics

Final project

Judith Sánchez¹ Stephani Vianey Francisco² Cristian Díaz³ Julio Ballinas⁴

FCFM | BUAP

Dr. Iván Fuentecilla Cárcamo

Puebla, 5th December, 2025



Contents

- 1 Introduction
- 2 Wave equation in dielectrics
- 3 TE polarization: E_y and H_x
- 4 Three-layer structure and boundary conditions
- 5 TE slab waveguide (Pochi Yeh)
- 6 Numerical TE mode patterns
- 7 Conclusion

Introduction

Motivation and goals

- We want to understand how **electromagnetic waves** propagate inside **dielectric media**
- Starting point: **Maxwell's equations** in a linear, homogeneous, source-free medium
- Then we specialize to **TE (Transverse Electric) waves** and to a simple planar structure
- Final goal: visualize the **guided TE modes** of a symmetric slab waveguide following Pochi Yeh (1988)

Wave equation in dielectrics

Maxwell equations in a dielectric

We consider a linear, homogeneous, isotropic, source-free medium

$$\rho_f = 0, \quad \vec{J}_f = 0$$

with constitutive relations

$$\vec{D} = \varepsilon \vec{E}, \quad \vec{B} = \mu \vec{H}$$

Then Maxwell's equations become

$$\nabla \cdot \vec{E} = 0$$

$$\nabla \cdot \vec{B} = 0$$

$$\nabla \times \vec{E} = -\mu \frac{\partial \vec{H}}{\partial t}$$

$$\nabla \times \vec{H} = \varepsilon \frac{\partial \vec{E}}{\partial t}$$

Derivation of the wave equation

Take the curl of Faraday's law

$$\nabla \times (\nabla \times \vec{E}) = -\mu \frac{\partial}{\partial t} (\nabla \times \vec{H})$$

Using Ampère–Maxwell

$$\nabla \times \vec{H} = \varepsilon \frac{\partial \vec{E}}{\partial t}$$

we get

$$\nabla \times (\nabla \times \vec{E}) = -\mu\varepsilon \frac{\partial^2 \vec{E}}{\partial t^2}$$

Vector identity with $\nabla \cdot \vec{E} = 0$

$$\nabla \times (\nabla \times \vec{E}) = -\nabla^2 \vec{E}$$

So the wave equation in a dielectric is

$$\nabla^2 \vec{E} - \mu\varepsilon \frac{\partial^2 \vec{E}}{\partial t^2} = 0$$

Derivation of the wave equation

and analogously for $\vec{H}$

$$\nabla^2 \vec{H} - \mu\epsilon \frac{\partial^2 \vec{H}}{\partial t^2} = 0$$

Plane wave solution and dispersion relation

Assume a plane wave of the form

$$\vec{E}(\vec{r}, t) = \vec{E}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)}$$

Substituting into the wave equation gives

$$\nabla^2 \vec{E} = -k^2 \vec{E}, \quad \frac{\partial^2 \vec{E}}{\partial t^2} = -\omega^2 \vec{E}$$

Then

$$(-k^2 + \mu\epsilon\omega^2)\vec{E} = 0$$

For non-trivial fields we obtain the [dispersion relation](#)

$$k^2 = \mu\epsilon\omega^2$$

So a plane wave is indeed a solution of the electromagnetic wave equation in a dielectric medium

TE polarization: E_y and H_x

TE polarization and field components

- TE (Transverse Electric) means that the electric field is perpendicular to the plane of incidence
- We consider propagation in the x - z plane, so the electric field is along y

$$\vec{E} = (0, E_y, 0), \quad \vec{H} = (H_x, 0, H_z)$$

We assume a plane wave

$$E_y(x, z, t) = E_{y0} e^{i(k_x x + k_z z - \omega t)}$$

Using Faraday's law $\nabla \times \vec{E} = -\mu \partial \vec{H} / \partial t$ we relate E_y and $\vec{H}$

Relations between E_y and $\vec{H}$

From the curl and time derivatives we obtain

$$(\nabla \times \vec{E})_x = \frac{\partial E_y}{\partial z} = ik_z E_y$$

$$(\nabla \times \vec{E})_z = -\frac{\partial E_y}{\partial x} = -ik_x E_y$$

Matching with $-\mu \partial \vec{H} / \partial t$ gives the [TE relations](#)

$$H_x = \frac{k_z}{\omega \mu} E_y, \quad H_z = -\frac{k_x}{\omega \mu} E_y$$

For normal incidence ($k_x = 0$) only H_x survives

$$H_x = \frac{k_z}{\omega \mu} E_y, \quad H_z = 0$$

These expressions are the basis for applying boundary conditions at interfaces

Three-layer structure and boundary conditions

Three-layer dielectric structure

We consider a planar structure:

- Medium I: $z < 0$, refractive index n_1
- Medium II: $0 < z < d$, refractive index n_2
- Medium III: $z > d$, refractive index n_1

For normal incidence ($k_x = 0$) and TE polarization

$$\begin{aligned} E_{x1} &= A e^{-ik_{z1}z} & z < 0 \\ E_{x2} &= B e^{-ik_z z} + C e^{ik_z z} & 0 < z < d \\ E_{x3} &= D e^{ik_{z1}z} & z > d \end{aligned}$$

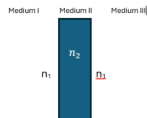


Figura: *

In each region the tangential electric field is a superposition of plane waves

$$E_y^{(j)}(z) = A_j e^{ik_{jz}z} + B_j e^{-ik_{jz}z}$$

with

$$k_{jz} = \omega \sqrt{\mu_j \epsilon_j}$$

Three-layer dielectric structure. Incident, internal and transmitted TE waves in a planar three-layer medium with refractive indices n_1 , n_2 , n_1 .

Jackson's boundary conditions

Jackson's boundary conditions (no surface charge, no surface current):

- Tangential $\vec{E}$ is continuous
- Tangential $\vec{H}$ is continuous

At $z = 0$:

$$E_y^{(I)} = E_y^{(II)}, \quad H_x^{(I)} = H_x^{(II)}$$

At $z = d$:

$$E_y^{(II)} = E_y^{(III)}, \quad H_x^{(II)} = H_x^{(III)}$$

Using $H_x = (k_z/\omega\mu)E_y$ this becomes a system of equations relating the amplitudes of incident, reflected, internal and transmitted waves. This is the core of Problems 3 and 4

TE slab waveguide (Pochi Yeh)

Symmetric slab waveguide

- High-index core with index n_2 and thickness d
- Two lower-index outer regions with index n_1
- Structure infinite in y and z , finite only in x

$$n_2 = 2,5, \quad n_1 = 1,5, \quad d = \lambda$$

Light is confined in the transverse x direction by [total internal reflection](#) at $x = \pm d/2$, and propagates along z as a guided mode

TE mode profiles and eigenvalue equations

The TE field is written as

$$E_y(x, z, t) = E_m(x) e^{i(\omega t - \beta z)}$$

Inside the core ($|x| < d/2$) the field oscillates, outside it decays exponentially. Two families:

Even TE modes

$$E_m^{\text{even}}(x) \sim \begin{cases} \cos(hx), & |x| < d/2 \\ \cos(hd/2) e^{-q(|x|-d/2)}, & |x| > d/2 \end{cases}$$

$$h \tan\left(\frac{hd}{2}\right) = q$$

Odd TE modes

$$E_m^{\text{odd}}(x) \sim \begin{cases} \sin(hx), & |x| < d/2 \\ \text{sgn}(x) \sin(hd/2) e^{-q(|x|-d/2)}, & |x| > d/2 \end{cases}$$

$$h \cot\left(\frac{hd}{2}\right) = -q$$

Effective indices for the first four TE modes

For the example considered by Yeh (1988) with $d = \lambda$ one obtains four guided TE modes

$$\bar{\beta}_m = \frac{\beta_m}{k_0} = \{2,463, 2,348, 2,148, 1,850\}, \quad m = 0, 1, 2, 3$$

Each effective index satisfies

$$n_1 < \bar{\beta}_m < n_2$$

and determines the transverse parameters

$$h_m = k_0 \sqrt{n_2^2 - \bar{\beta}_m^2}, \quad q_m = k_0 \sqrt{\bar{\beta}_m^2 - n_1^2}$$

These values are the input for the numerical simulation of the field patterns in the slab

Numerical TE mode patterns

Numerical simulation setup

- Python + matplotlib used to compute

$$\text{Re}[E_y^{(m)}(x, z, t_0)] = E_m(x) \cos(\beta_m z)$$

at a fixed time t_0

- Grid in the (x, z) plane:

$$x \in [-d, d], \quad z \in [0, 3d]$$

- Custom colormap based on the project color palette
- Same color scale for all modes to allow direct comparison

TE slab modes: field patterns

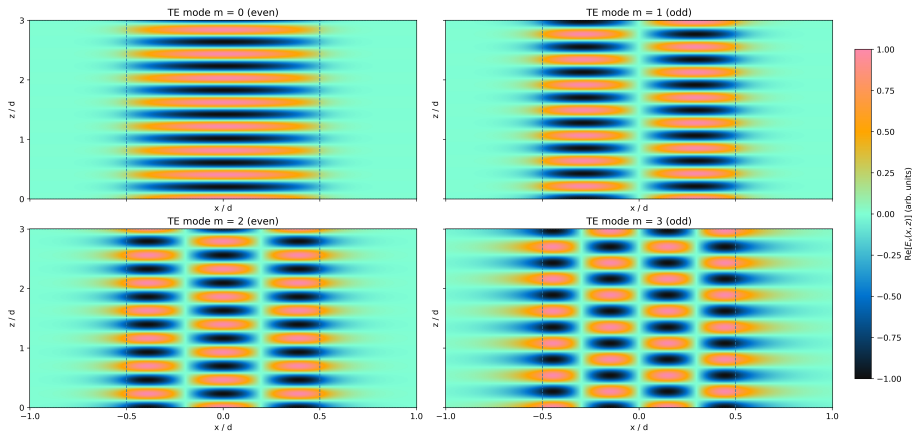


Figura: *

Real part of the electric field $\text{Re}[E_y(x, z)]$ for the first four TE guided modes in a symmetric slab waveguide ($n_2 = 2,5$, $n_1 = 1,5$, $d = \lambda$)

Reading the figure: confinement and symmetry

- Dashed vertical lines mark the **core boundaries** at $x = \pm d/2$
- Warm colors (orange–pink) represent **positive** field values, dark blue–black represent **negative** values
- As the mode index m increases, more **nodes** appear in the transverse direction x
- Even modes are **symmetric** with respect to $x = 0$, odd modes are **antisymmetric**, changing sign across the center
- Along z the pattern oscillates with $\cos(\beta_m z)$, showing the periodic variation of the field as the wave propagates

Conclusion

Conclusion

- Starting from Maxwell's equations in a dielectric we derived the **wave equation** and the **dispersion relation** for plane waves
- For TE polarization we obtained explicit relations between the transverse electric field E_y and the magnetic components H_x and H_z
- Using Jackson's boundary conditions we analyzed a three-layer structure and wrote the system that couples the amplitudes of incident, reflected and transmitted waves
- Following Pochi Yeh we studied a symmetric slab waveguide and its **even/odd TE modes**, determined by simple eigenvalue equations
- Finally, we computed and visualized the real part of $E_y(x, z)$ for the first four TE guided modes, showing **confinement**, **symmetry**, and the increasing complexity of higher-order modes

Thank you!